

! compare analog and digital signals

Transmission Basics

- Modern networks use many different and advanced transmission methods.
- This part explains the key features of how data is sent today.

Analog Signaling

- Analog signals **change continuously and smoothly**, like a wave.
- They have four main properties:
 - **Amplitude(Genlik)**: how strong the signal is at a moment
 - **Frequency**: how often it repeats in a time period
 - **Wavelength**: the distance between two wave peaks
 - **Phase**: the position of the wave compared to a fixed point in time.

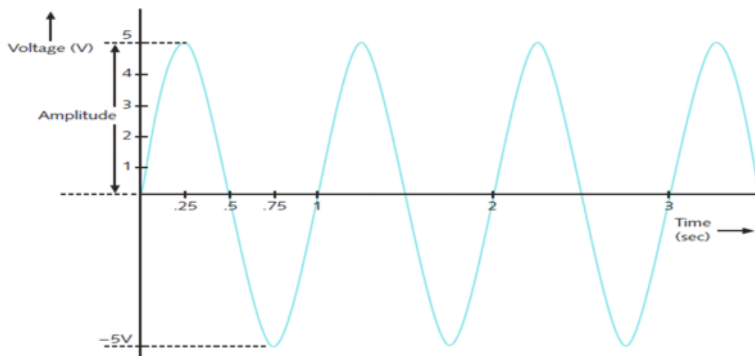


Figure 5-1 An example of an analog signal

- A main drawback of analog signals is their **varied and imprecise voltage**, which can cause transmission problems.
- Analog signals are **easily affected by noise or interference**, which can cause **errors during data transmission**.

Pulses of voltages ;

- Positive voltage represents a 1
- Zero voltage represents a 0

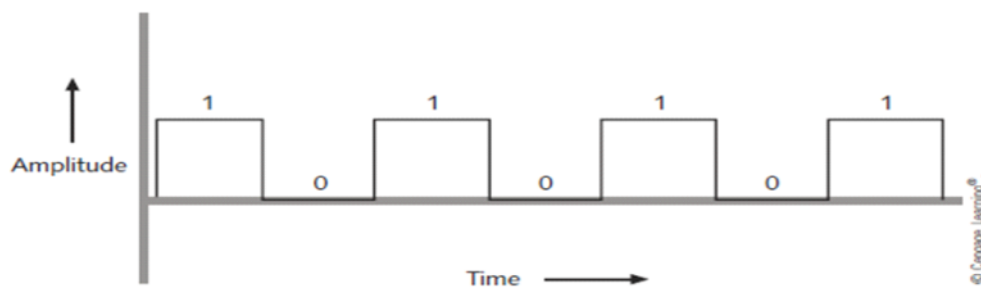


Figure 5-3 An example of a digital signal

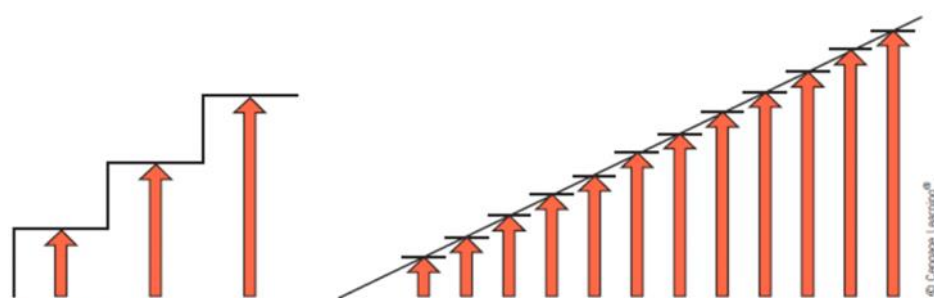


Figure 5-4 It's impossible to measure every height along a ramp

digital signals

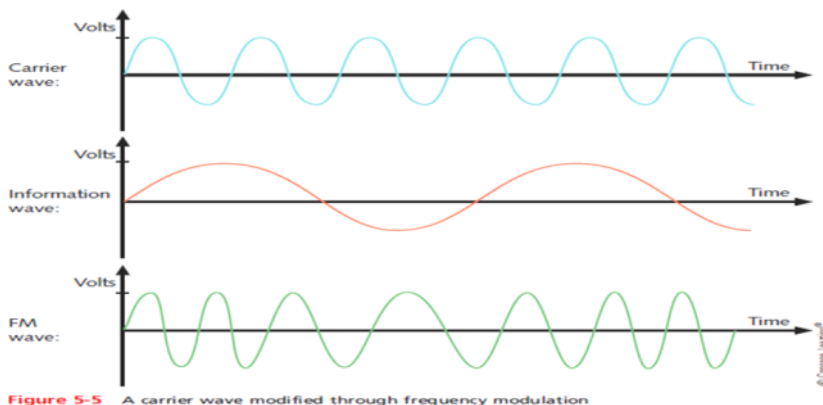
analog signals

Digital signals have defined levels and are **easier to measure**, while analog signals vary continuously and are **more complex to measure** accurately.

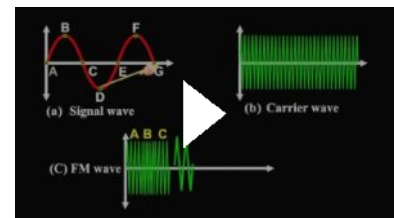
Data Modulation

- **Data relies on digital transmission**
- **Network connection may handle only analog signals**
- **Modem**
 - Translates between analog and digital signals
 - Stands for modulator/demodulator
- **Data modulation**
 - Changes analog signals into digital and vice versa
 - Makes analog signals ready to carry data across a network
- **Carrier wave**
 - Mixed with another analog signal
 - Creates a unique signal
 - Sent from one device to another
 - Has fixed properties
 - Purpose : Used to carry information
- **Information wave (data wave)**
 - Added to the carrier wave
 - Changes one of the carrier wave's properties
- **Frequency modulation**
 - The frequency of the carrier changes with the data signal
- **Amplitude modulation**
 - The amplitude of the carrier changes with the data signal
- **Digital Subscriber Line (DSL)**
 - Uses modulation for data transmission

carrier = taşıyıcı



Amplitude Modulation and Frequency Modulation





Baseband and Broadband Transmission

- **Baseband transmission:**
 - Sends digital signals on one channel.
 - Uses **all the cable's capacity** for a single signal.
 - Example: Ethernet.
- **Broadband transmission:**
 - Sends multiple signals on the same medium.
 - Uses multiplexing to manage different signals.
 - Example: Cable TV and Internet.

Multiplexing

- **Multiplexing:** A method that lets many signals travel at the same time over one medium.
- **Subchannels:** Smaller logical channels created from one main channel.
- **Multiplexer (mux):**
 - Combines signals from many channels.
 - Needed on the sender side.
- **Demultiplexer (demux):**
 - Splits the combined signals at the receiving side.

TDM (Time division multiplexing)

Divides channel into multiple time intervals

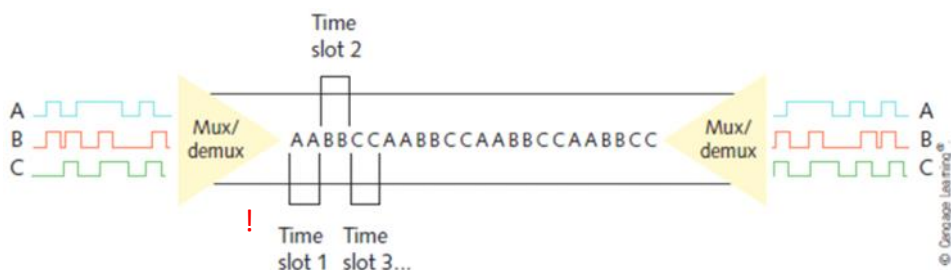


Figure 5-6 Time division multiplexing

- TDM sends signals from different sources one by one using small time parts.
- The signals are combined, sent together, and then separated again.
- This lets many signals use the same channel without mixing.

Statistical multiplexing

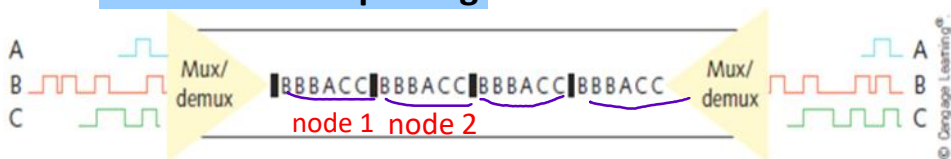


Figure 5-7 Statistical multiplexing

- Transmitter assigns slots to nodes
- According to priority and need
- More efficient than TDM

FDM (Frequency Division Multiplexing)

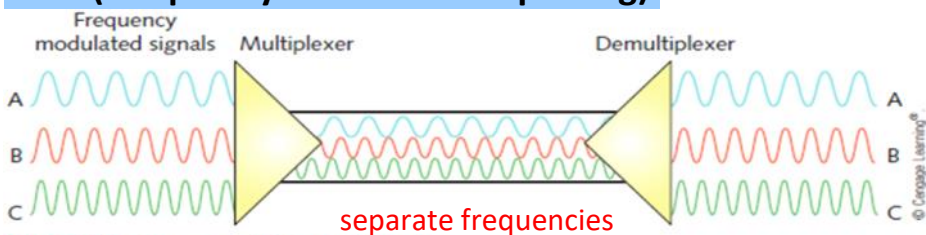


Figure 5-8 Frequency division multiplexing

- FDM gives each communications subchannel a unique frequency band.
- Signals are sent at the same time but on separate frequencies.
- Examples : mobile phone networks and DSL internet.

! WDM (Wavelength Division Multiplexing)

Wavelength division multiplexer

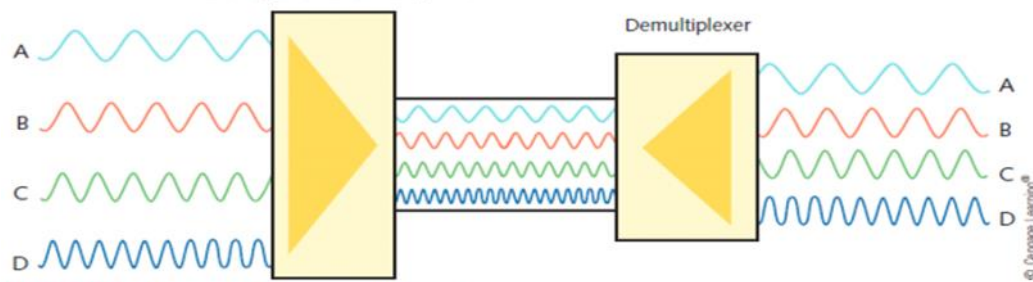


Figure 5-9 Wavelength division multiplexing

- Uses one **fiber-optic cable** for transmission.
- **Sends multiple light signals at the same time** simultaneously using different wavelengths.

Throughput and Bandwidth

• Throughput

- Amount of data sent transmitted during a given time.
- Also called effective data rate or payload rate.
- Measured in bits transmitted per second.

• Bandwidth

- Difference between highest and lowest frequencies.
- Shows the range of frequencies.

• Note

- Both are measured as bits transmitted per second, known as bit rate.

Twisted-Pair Cable

- Made of color-coded insulated copper wire pairs. !
- Each wire is **0.4 to 0.8 mm** in diameter and covered with **plastic insulation**. !
- Every two wires are twisted together.

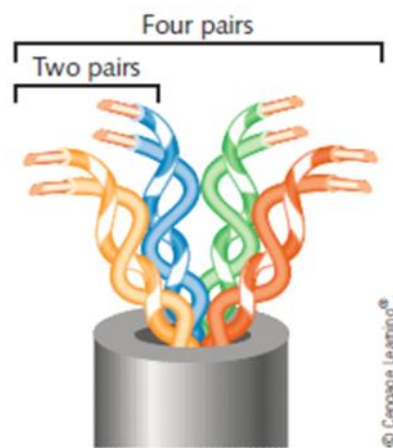


Figure 5-13 Twisted-pair cable

- More wire pair twists per foot: (Ayak başına daha fazla kablo bükümü)
 - Better protection against crosstalk
 - Higher signal quality
 - Higher cost
- Twist ratio: (büküm oranı)
 - Number of twists per meter or foot
- High twist ratio: (yüksek büküm oranı)
 - Can cause more signal loss (attenuation)
- **Many design options:**
 - Based on twist ratio, number of wire pairs, copper quality, and shielding
 - Can include from 1 to 4200 wire pairs
- **Wiring standards:**
 - Follows TIA/EIA 568 specification
- **Common cable types:**

fazla bilgi-gereksiz kasmayın

- Categories: CAT 3, 5, 5e, 6, 6a, 7 Modern LANs usually use CAT 5e or higher

Twisted-Pair Cable

- **Advantages:**

- Affordable and easy to install
- Flexible and supports different network layouts
- Can cover long distances without a repeater

- **Two categories:**

- **STP** – Shielded Twisted Pair
- **UTP** – Unshielded Twisted Pair

STP (Shielded Twisted Pair)

- Each wire is covered with **insulation(yalıtım)**.
- The cable is wrapped in metallic shielding (like foil):
 - This **blocks electromagnetic interference** from outside.
 - Keeps the electrical signal **energy inside**.
 - It can also be **grounded** for extra safety.

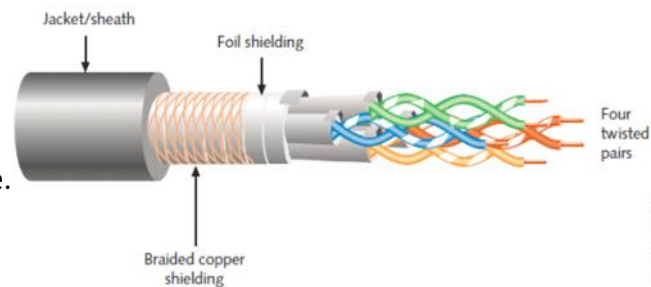


Figure 5-14 STP cable

UTP (Unshielded Twisted Pair)

- Contains one or more insulated wire pairs.
- Covered with a plastic sheath.
- No extra shielding:
 - It is cheaper but less noise resistance

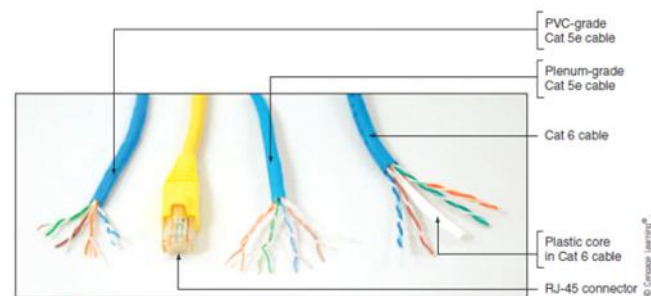


Figure 5-15 Various UTP cables and RJ-45 connector

STP vs UTP – Comparing

- **Throughput:**

Both STP and UTP can send data at the same speed.

- **Cost:**

STP is usually more expensive than UTP.

- **Connector:**

Both use the same connector type – RJ-45.

- **Noise Immunity:**

STP has better protection against electrical noise.

- **Size and Scalability:**

Both support up to 100 meters of cable length.

Cable Pinouts – Summary

- **Straight-through cable**
RJ-45 connectors are the same on both ends.
- **Crossover cable**
Transmit and receive wires are switched on one end.
- **Rollover cable**
All wires are reversed and mirrored.
Also known as Yost or Cisco console cables.

Used to connect a computer to a router's console port.

Fiber-Optic Cable

- **Fiber-optic cable:**
Has one or more glass or plastic fibers in the center (core).
- **Data transmission:**
Sends pulsing light from lasers or LEDs through the core to carry data.
- **Cladding(Kaplama):**
 - A glass or plastic layer around the fibers.
 - Has a different density to keep the light inside by reflecting it.
 - Helps the fiber bend without losing the signal.
- **Plastic buffer(tampon) around the cladding**
Protects the cladding and core, blocks escaping light, and is surrounded by Kevlar fiber strands for extra strength.
- **Plastic sheath(dış kaplama)**
Covers the Kevlar strands and holds the cable together.
- **Advantages over copper cables**
 - Offers very **high data speed** (throughput)
 - Strong protection against **noise**
 - Provides excellent **security**
 - Can send signals **over long distances**
 - Used widely for **fast** networks
- **Disadvantages**
 - More **costly** than twisted pair cables
 - Needs special tools for joining (splicing)

! adv. of fiber optic cable and know type of them

Fiber-Optic Cable Summary

- **Throughput(veri hacmi)**
 - Can transmit data up to 100 Gbps per channel reliably.
- **Cost**
 - Most expensive type of transmission cable.
- **Noise Immunity**
 - Not affected by electromagnetic interference (EMI).
- **Size and Scalability**
 - Cable lengths range from 150 to 40,000 meters.
 - Depends on the wavelength of light and cable type.

Single Mode Fiber (SMF)

- **Narrow Core:**
SMF has a very thin core (8–10 microns), Light travels in one **straight** path.
- **Long-Distance Transmission:**
It can send signals over long distances without needing repeaters.
- **Not for Short Distances:**
It's rarely used for short links because it's expensive.
However, it's used in the Internet backbone.

Multimode Fiber(MMF)

- **Larger Core Size:** Has a wider core than single-mode fiber, usually 50 or 62.5 microns.
- **Light Travel:** Light from a laser or LED travels at **different angles** inside the core.
- **Higher Attenuation:** It has **more signal loss than single-mode fiber**.
- **Typical Uses:**
 - Connecting routers to switches
 - Linking servers to the network backbone

Summary

- Computers use electrical current to send and understand data as binary (on/off signals).
- A **channel** is a communication path between devices, either physical or logical.
- **Baseband** allows one transmission at a time, while **broadband** supports many transmissions using **multiplexing**.
- **Throughput** shows how much data is sent in a certain amount of time.

📊 Analog vs Digital Signals – Comparison Table			📄
Feature	Analog Signal	Digital Signal	
Waveform	Continuous and smooth wave (like a sine wave)	Square wave or pulses (on/off, 1/0)	
Values	Infinite possible values	Only two levels: 0 and 1	
Amplitude & Voltage	Changes continuously (varied and imprecise)	Fixed levels (e.g., 0V and 5V)	
Measurement	Harder to measure accurately	Easier to measure and process	
Noise Resistance	Easily affected by noise or interference	Less affected by noise (more reliable)	
Error Possibility	High – due to continuous variation and noise	Low – clearly defined binary states	
Transmission Type	Real-world signals (e.g., sound, light)	Used in computers and digital communication	
Complexity	More complex and harder to filter	Simpler and easier to handle	